Beyond the Blob: Localizing Inserted Lesions with Aligned Vessel Forecasts

Anonymous CVPR submission

Abstract

Pulmonary vessels that bend or branch inside a candidate region of a chest CT scan often produce compact responses that resemble small nodules. A candidate scorer may then accept the region for the wrong reason, or return a location on the vessel rather than on the lesion. A natural remedy is to forecast the vessels from the surrounding tissue and to account for them before scoring, yet it remains unclear whether such a forecast helps because it is correctly placed or merely because it supplies additional context. We isolate this question with paired CT crops that are identical outside a 14-mm ball and differ only by an inserted lesion, so that a forecast fitted to the exterior is shared by both. A 90° *rotation of the same forecast, which preserves its values but breaks its alignment with the anatomy, serves as a sham. At a matched 10% candidate false-positive rate (FPR), the aligned forecast places the returned location within 3 mm of the lesion for* 6.09 *percentage points (pp) more insertions than the rotated one (95% CI* $[+4.38, +7.78]$*) on a held-out LUNA16 set of 170 patients and 12,240 insertions, and for* 2.76 pp *(*$[+1.77, +3.76]$*) more on 72 HiPaS scans with manually annotated native vessels. On a low-texture LUNA16 subset, alignment also increases crop-level sensitivity by* 9.37 pp *(*$[+4.56, +12.29]$*), whereas on the broader held-out set the crop-level gain is smaller and its interval includes zero. Subtracting the forecast at a fixed amplitude localizes as many lesions as estimating the amplitude jointly with the lesion. Correct placement of a vessel forecast therefore mainly changes the location returned by a candidate scorer, at an unchanged false-alarm budget.*

1. Introduction

Candidate scorers for pulmonary nodule detection respond to compact, approximately spherical bright structures in chest CT. Many such responses are produced by vessels: a vessel imaged in cross-section, a bend, or a junction all yield compact peaks (Fig. 1a). A scorer must therefore make two related decisions, namely whether a candidate crop contains a lesion and where within the crop the lesion lies. Vessels interfere with both. They can raise the score of a lesion-free crop, and within a crop that does contain a lesion they can produce the maximum of the spatial search, so that the location presented to a reader falls on the vessel.

Because vessels that traverse a candidate region are usually visible where they enter and leave it, their course through the region can be forecast from the surrounding tissue and accounted for before the search (Fig. 1c). Demonstrating that such a forecast helps for the intended reason is less straightforward than it may appear. A forecast adds input and model capacity, and it correlates with the type of background in the crop; each of these effects could raise a score even if the forecast were spatially misplaced.

In this work we test a specific hypothesis: a vessel forecast improves candidate scoring *because it is aligned with the anatomy*. The experimental design rests on two controls. First, each control crop and its lesion-bearing counterpart are identical outside a 14-mm ball that contains the lesion, so a forecaster that observes only the exterior produces the same forecast for both (Fig. 1b). Second, a sham forecast is obtained by rotating the same field by a quarter-turn about the slice axis; the sham retains every value, scale and degree of smoothness of the forecast but no longer coincides with the vessels (Fig. 1c). A third control supplies the classifier with summary statistics of the exterior only. We further examine how much flexibility the vessel estimate requires by evaluating a single local scoring function, which we call a *readout*, under four treatments of the vessel amplitude ranging from no forecast to a joint fit with the lesion (Sec. 3.3).

Our contributions are as follows:

- a paired intervention that separates the effect of forecast *placement* from the information a forecast carries, evaluated at a fixed candidate false-positive budget;
- a comparison of readouts under a shared search and classifier, showing that subtracting the forecast at a fixed amplitude performs as well as estimating the amplitude jointly with the lesion;
- a localization analysis on a held-out LUNA16 set and on an independent dataset with annotated native vessels, showing that alignment mainly changes the location returned by the scorer rather than the number of crops it accepts.

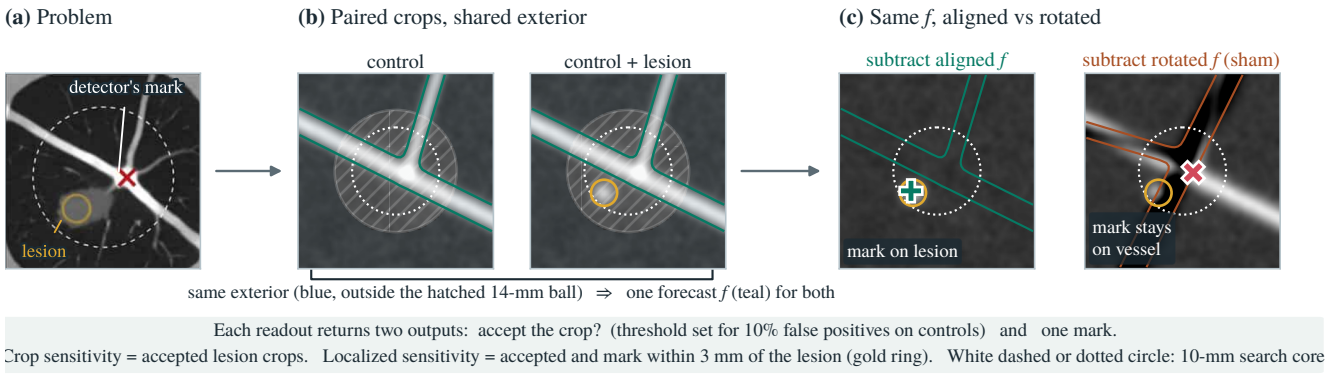


Figure 1. **Problem and approach.** (a) A compact-blob search inside the 10-mm core returns the bright vessel junction (red cross) instead of the nearby lesion (gold ring). (b) A control crop and its lesion-bearing counterpart are identical outside 14 mm. The forecaster observes only this exterior and continues the vessels across the hidden ball, so both crops receive the same forecast f. (c) Subtracting the aligned forecast removes the vessel, and the search returns the lesion (green cross). The rotated sham has the same values in the wrong place; the vessel remains and still yields the returned location. Panel (a) was created in BioRender. Panels (b) and (c) use a generated slice with the same layout, and their returned locations are computed with a simplified version of the readout in Sec. 3.3.

2. Related Work

Vessels in nodule candidate scoring. Multiscale Hessian filters characterize local curvature and are widely used to separate tubular from compact structures [5, 9, 12]. Junctions nevertheless produce curvature along several axes, and the strongest response in a search region may lie away from the lesion. Agam *et al*. [1] used reconstructed vessel trees to resolve such local ambiguities in CT, and Charbonnier *et al*. [3] used vessels outside a subsolid nodule to guide vessel segmentation within it. Learned candidate systems exploit local anatomy or multiple views [14, 16], and Gu *et al*. [7] added a vessel-suppression branch to a nodule detector. These works motivate the use of vascular context; we instead hold the exterior signal fixed and vary only its spatial alignment with the candidate.

Lesion insertion for controlled evaluation. Inserting simulated lesions into patient CT produces hybrid images in which the location, size and morphology of the target are known. Robins *et al*. [11] compared projection- and image-domain techniques for virtual lung nodule insertion and found both to yield realistic renderings, and Gong *et al*. [6] proposed learned lesion and noise insertion for virtual clinical trials in chest CT. We use image-domain insertion not to emulate clinical prevalence but as a paired intervention: control and lesion crops share every exterior voxel, which fixes the forecast and permits a placement sham.

Task-based assessment of image operations. Improvements in pixel-level fidelity need not translate into better detection. Li *et al*. [8] compared denoising methods across detection observers, and Yu *et al*. [17] reported that image-quality measures and detection efficiency can move in opposite directions. The operation studied here is vessel subtraction or conditioning, which we assess by sensitivity at a fixed false-positive budget and by whether the returned location coincides with the inserted target.

Detection with structured interference. Detecting a non-negative signal in the presence of a structured interference subspace is a classical problem in signal processing [13]. Our readouts use a weighted affine background model and a single exterior-derived nuisance component; the comparison concerns the origin of that component, the treatment of its amplitude, and the resulting effect on a searched candidate score.

3. Method

3.1. Problem Setup and Paired Insertions

Each input is a CT crop y of 64^3 voxels with spacing $1.5 \times 0.7 \times 0.7$ mm (slice, row, column), centered on a candidate proposal c. We distinguish three nested regions: the search *core* within 10 mm of c, the *hidden ball* within 14 mm, and the *exterior* E beyond 14 mm, whose inner part between 15.5 and 19.5 mm we call the *shell*. A scorer consists of a readout, which maps the crop to template scores, and a *classifier* on top of them. It produces a crop decision $A(y) \in \{0, 1\}$, which accepts the crop for review, and a location $L(y)$, the single point it returns.

Let b denote a CT background, v an optional added vessel field, and s an inserted lesion with $s = 0$ on E. The control

and lesion crops are

$$y_0 = b + v, \qquad y_1 = b + v + s. \tag{1}$$

For native-vessel backgrounds we set $v = 0$, since the vessels are already part of b. Because y_0 and y_1 coincide on E, the forecast $f = F(y|_E)$ is identical for a control and all of its insertions. All scorers evaluate y_0 and y_1 at the same center c and may use the entire crop; only the forecaster is restricted to E. A score can thus increase because a lesion is present, but not because the lesion altered the forecast.

Placement sham and exterior-summary control. Let Q denote a 90° rotation about the slice axis through c. Since the row and column spacings are equal, Q permutes voxels without interpolation. The sham forecast Qf preserves the radial profile, the multiset of values, the smoothness and the shape of f, but in general moves its vessels to locations without vascular anatomy. We refer to f and Qf as the *aligned* and *rotated* placements. Each aligned scorer is compared with (i) the rotated placement with a refitted classifier, which allows the sham to make the best possible use of a misplaced field, and (ii) the rotated placement evaluated with the aligned classifier, which removes any effect of refitting. A third control, denoted CT + shell, receives four summary statistics of the shell but no spatial field. Gains over all three controls are attributed to placement.

3.2. Exterior Vessel Forecast

The forecaster continues vessels whose entry and exit points are visible in the exterior. Peaks on a sphere of radius $15.5\,\mathrm{mm}$ propose pairs of entry and exit points, and a pair is retained if the signal continues against its transverse flanks at radii of 17.8 and $20.4\,\mathrm{mm}$ on both sides. For each retained chord we fit a Gaussian cross-section, with width and gain, to the two exterior segments, interpolate width and amplitude along the chord, and taper the field smoothly beyond $22\,\mathrm{mm}$. The resulting forecast f is an additive estimate in CT units; a crop without an accepted chord has $f = 0$ and is processed by the same procedure. All forecaster constants are listed in Sec. S1 of the supplement. We verify the shared-exterior property by confirming that the rotation preserves the value multiset of every control, that rerunning the forecaster after replacing the hidden region yields an identical field, and that the visible exteriors of every rendered pair agree bitwise. A straight chord is an approximation that curved paths and branches can violate; the native-vessel experiment in Sec. 5.5 evaluates this case directly.

3.3. Forecast-Conditioned Readouts

For each search center x in the core, let W_x be Gaussian weights ($\sigma = 3.4\,\mathrm{mm}$) on a $6\,\mathrm{mm}$ ball with inner product $\langle a, b\rangle_W = \sum_p W_x(p)\, a(p)\, b(p)$, and let B span a constant and the three spatial coordinates within the window. Projecting the observation, the forecast and a compact template onto the W_x-orthogonal complement of B gives

$$z = P_x y, \qquad u = P_x f, \qquad t = P_x T_{x,r}, \tag{2}$$

with templates $T_{x,r}(p) = [1 - \|p - x\|^2/r^2]_+^{0.8}$ and radii $r \in \{1.85, 2.7, 3.7\}\,\mathrm{mm}$. The projection removes the local background level and linear drift. We model the projected observation as

$$z \approx g\,u + h\,t, \qquad h \geq 0, \tag{3}$$

where g is the vessel gain and h the lesion amplitude. In the absence of a lesion ($h = 0$), the gain is estimated with a penalty toward the exterior value $g = 1$,

$$J_0 = \min_g \|z - gu\|_W^2 + \lambda(g-1)^2, \qquad \lambda = \rho\|u\|_W^2, \tag{4}$$

whose minimizer is $g_0 = (\langle u, z\rangle_W + \lambda)/G$ with $G = \|u\|_W^2 + \lambda$.

Treatment of the vessel gain. Adding the lesion term reduces the loss by $[a]_+^2/d$, where a is the correlation of the residual with the template and d an effective template energy. The four readouts differ only in how a and d are formed. The *CT-only* readout ignores the forecast ($u = 0$), so that $a = \langle t, z\rangle_W$ and $d = \|t\|_W^2$. *Fixed subtraction* keeps the exterior gain $g = 1$, giving $a = \langle t, z - u\rangle_W$, $d = \|t\|_W^2$ and $J_0 = \|z - u\|_W^2$. The *plug-in* readout holds the no-lesion estimate g_0 fixed, so that $a = \langle t, z\rangle_W - g_0\langle t, u\rangle_W$ and $d = \|t\|_W^2$. Finally, the *joint profile* (profile for short) fits g and h together; eliminating g yields

$$J(h) = J_0 - 2ah + h^2 d_{\mathrm{prof}}, \qquad d_{\mathrm{prof}} = \|t\|_W^2 - \frac{\langle t, u\rangle_W^2}{G}, \tag{5}$$

with the same a as the plug-in readout. Profile and plug-in coincide when template and forecast are orthogonal after projection; where they overlap, $d_{\mathrm{prof}} < d$ and profiling assigns more weight to the same residual.

The treatment of the gain matters most at the lesion itself. For $z = z_0 + h_\star t$, the plug-in gain changes by $h_\star\langle u, t\rangle_W/G$, so the plug-in correlation increases by only $h_\star(\|t\|_W^2 - \langle u, t\rangle_W^2/G)$, whereas the correlation under fixed subtraction increases by the full $h_\star\|t\|_W^2$. A gain estimated inside the core can thus absorb part of the lesion, while a fixed gain cannot but inherits any amplitude error of the exterior fit. For an empty forecast the four readouts reduce to the same local test, so differences on such crops arise only from the fitted classifier coefficients.

3.4. Scoring, Search and Classification

At each center and radius the normalized score is

$$m_r(x) = \frac{[a]_+/\sqrt{d}}{\sigma_0}, \qquad \sigma_0 = \sqrt{J_0/\textstyle\sum_p W_x(p) + \tau^2}, \tag{6}$$

Algorithm 1 Forecast-conditioned candidate scoring

Require: crop y, center c, readout $\in$ {CT-only, fixed, plug-in, profile}
$f \leftarrow F(y|_E)$ ▷ forecast from voxels beyond 14 mm
for centers x within 10 mm of c, radii $r \in \{1.85, 2.7, 3.7\}$ mm **do**
  $z, u, t \leftarrow P_x y,\ P_x f,\ P_x T_{x,r}$ ▷ affine residuals
  $J_0, g_0 \leftarrow$ no-lesion fit, Eq. (4) (fixed: $g = 1$)
  $a, d \leftarrow$ correlation and energy for the readout
  $m_r(x) \leftarrow [a]_+ / \left(\sqrt{d}\,\sigma_0\right)$ ▷ Eq. (6)
end for
$\mathbf{q} \leftarrow$ shared CT features $\oplus\, (\max_x m_r(x))_r$
$A(y) \leftarrow [\mathrm{logit}(\mathbf{q}) > \kappa]$ ▷ κ at 10% FPR
$L(y) \leftarrow \arg\max_{x,r} m_r(x)$
return $A(y)$, $L(y)$

Role	Blocks	Controls	Insertions
Focused fit	46	138	2,484
Focused development	39	117	2,106
Focused calibration	92	276	–
Focused evaluated	183	549	9,882
Broader fit	80	240	4,320
Broader development	85	255	4,590
Broader calibration	175	700	–
Broader holdout	170	680	12,240
HiPaS development	24	144	2,592
HiPaS transfer	72	432	7,776

Table 1. **Cohorts.** Blocks are LUNA16 patients or HiPaS scans. Each control contributes one negative decision and is paired with 18 insertions.

i.e., the square root of the loss reduction scaled by the readout's own null residual with floor τ. The per-radius maxima $\max_x m_r(x)$ are passed to a ridge-logistic classifier, and the overall maximizer over x and r defines the location $L(y)$ (Algorithm 1). All classifiers share 12 local intensity and Hessian summaries, three template maxima from a larger window and three CT-only maxima. The CT + shell control adds its four shell summaries, and each forecast-conditioned readout adds its three conditioned maxima, so that fixed subtraction, plug-in and profile have identical feature counts and regularization budgets. Each readout is tuned over three coupled settings of the gain penalty ρ, the floor τ and the ridge strength (Sec. S2).

3.5. Baselines

The Hessian baseline subtracts the same forecast at a strength of 0.35, 0.7 or 1.0 before computing a standard compactness-curvature statistic at Gaussian scales of 1.3, 2.5 and 4.0 mm. It measures the contribution of a different readout when the nuisance estimate is held fixed. The learned baseline is a hybrid CNN with a small 3D feature extractor and a shared CT feature path, whose second input is either smoothed CT (the CT-only CNN), the aligned forecast, the rotated forecast or the residual $y - f$; architecture, data, seeds and training budget are identical across inputs (Sec. S6).

4. Experimental Setup

4.1. Datasets

We evaluate on three settings derived from LUNA16 [2, 15] and HiPaS [4], summarized in Table 1. In each setting a *block* is a patient (LUNA16) or a scan (HiPaS), and every control crop is paired with a bank of 18 insertions that vary lesion shape, contrast and offset.

Low-texture LUNA16 subset. This subset comprises 549 neighborhoods from 183 patients, each with a single, crossing or skew added vessel. The core intensity standard deviation is capped at 42 HU to isolate cases in which vessels compete strongly with the lesion, and the lesions are compact radially symmetric signals. Thresholds and the analysis for this subset were fixed before testing. Because these cases also informed the design of the shared readouts, we treat readout comparisons on them as development results.

Broader LUNA16 held-out set. The held-out set uses 170 patients disjoint from all fitting, development and calibration data, with four parenchymal neighborhoods per patient (680 controls and 12,240 insertions). It imposes no texture cap (median core SD 117 HU), uses added vessels that bend and taper, and introduces bilobed and part-solid-like lesions at three contrasts (84, 196 and 364 HU) and three offsets (0, 4 and 7 mm) from the proposal center. These shapes differ from the search templates. Sampling constraints and the lesion model are detailed in Sec. S3.

HiPaS native vessels. The native-vessel set consists of 72 noncontrast scans from a different source with manual artery and vein annotations. Six neighborhoods per scan are centered between nearby annotated arteries and veins, yielding 432 controls and 7,776 insertions without added vessels (median core SD 253 HU). The annotations define only the sampling population and the analysis strata; the scorer observes the image alone.

4.2. Model Selection and Evaluation

Prespecified analysis. Classifiers for the broader set are fitted on 80 patients and tuned on 85 others. Before the test images were rendered, we fixed the cohort, the insertion bank, the analysis plan and three scorers selected on development: the aligned profile as the lead scorer, fixed subtraction as the strongest alternative readout, and the rotated profile with a refitted classifier. The primary criterion was a crop-sensitivity gain of at least 5 pp at 10% FPR, with a

positive lower confidence limit against both the rotation and fixed subtraction; localized sensitivity was a prespecified secondary endpoint. A development bootstrap indicated that approximately 123 patients provide 80% power to detect a true 5-pp gain over the rotation, and 74 patients over fixed subtraction.

Operating points. We report two operating points. The *matched* operating point sets each scorer to a 10% candidate FPR on the evaluation controls via the order statistic of their scores and therefore measures ranking at a fixed budget. The *frozen* operating point applies cutoffs derived from 700 independent negative crops of 175 LUNA16 calibration patients, accepting scores strictly above the 10% order statistic; its achieved FPR may differ between scorers.

Metrics. Crop sensitivity is the fraction of lesion crops accepted. Localized sensitivity additionally requires the returned location to lie within 3 mm of the insertion center $x_\star$,

$$\Pr\big(A(y)=1,\ \|L(y)-x_\star\|\le 3\,\mathrm{mm}\mid Y=1\big). \quad (7)$$

Controls count against the FPR budget irrespective of their returned location.

Uncertainty. We draw 3,000 bootstrap resamples of blocks, stratified by release fold, retaining each block's full insertion bank and controls so that all comparisons are paired. For matched intervals the cutoff is recomputed in every draw. For frozen cutoffs we report intervals conditional on the calibration set and, separately, intervals that also resample the calibration patients. All differences are reported in absolute percentage points with 95% confidence intervals.

5. Results

5.1. Effect of Forecast Alignment

Low-texture subset. On the low-texture subset, the aligned profile outperformed its rotation by $+9.66$ pp [+8.10,+11.19] at frozen cutoffs (achieved FPR 10.38% *vs.* 10.02%) and by $+9.37$ pp [+4.56,+12.29] at matched FPR (Fig. 2b). Under the shared readouts, fixed subtraction gained $+14.72$ pp [+11.11,+17.05] over its rotation and $+14.15$ pp [+10.73,+16.44] over the CT + shell control, gaining 2,001 and losing 546 positive decisions relative to the rotation (Table 2).

Broader held-out set. The broader set is considerably harder, with crop sensitivities of 21–24% for all scorers. The aligned profile increased crop sensitivity by $+2.08$ pp [-0.27,+4.40] over the rotation with a refitted classifier and by $+4.05$ pp [+1.37,+6.23] over the rotation evaluated with the aligned classifier. At frozen cutoffs the difference was $+4.50$ pp, albeit at achieved FPRs of 10.59% *vs.* 8.82%; including calibration uncertainty, the interval was [-0.59,+8.31] pp and the upper limit on excess FPR 5.00 pp. The prespecified crop-level criterion was therefore not met. Localization, in contrast, improved substantially (Fig. 3): the rotated profile returned the lesion location for 7.35% of insertions and the aligned profile for 13.45%, a gain of $+6.09$ pp [+4.38,+7.78].

Readout	Low-text. sens.	Broader held-out set sens.	loc. sens.	frozen
Fixed, aligned	50.75	24.05	15.38	24.74 / 10.59
Fixed, rotated	36.03	20.81	5.72	21.18 / 10.15
Plug-in, aligned	39.45	22.54	11.54	23.86 / 10.44
Plug-in, rotated	37.39	21.26	6.92	19.81 / 9.12
Profile, aligned	**46.50**	**23.32**	**13.45**	**23.79 / 10.59**
Profile, rotated	37.13	21.23	7.35	19.29 / 8.82
Prof. rot., aligned clf.	–	19.26	9.19	19.20 / 10.00
CT + shell	36.60	21.50	13.49	21.66 / 10.29
CT only	32.88	21.39	13.37	23.78 / 11.62
Hessian subtr.	47.34	14.31	–	14.31 / 10.29
Lead minus … (pp, 95% CI)				
− refitted rotation	–	+2.1 [−0.3, +4.4]	+6.1 [+4.4, +7.8]	–
− fixed subtr.	–	−0.7 [−3.0, +0.5]	−1.9 [−3.3, −1.0]	–

Table 2. **Placement and readout comparisons.** Sensitivity (%) at the matched 10% FPR on the low-texture subset (development) and the broader held-out set. Localized sensitivity (loc. sens.) requires acceptance and a returned location within 3 mm (Eq. (7)). The frozen column gives sensitivity / achieved FPR at the independent LUNA16 cutoffs. Bold: lead scorer selected before testing.

5.2. Acceptance versus Localization

A crop-by-crop comparison of the two placements explains the difference between the endpoints. Of the net gain of 746 localized insertions, about a third arises from crops accepted under both placements, where alignment moves the returned location onto the lesion (386 crops, against 124 moved away); the remainder comes from crops that only the aligned profile accepts (647 *vs.* 163; Table S2). Localization therefore improves considerably more than acceptance. This behavior follows from the competition illustrated in Fig. 1: if a vessel response $m(x_v)$ exceeds a lesion response $m(x_\ell)$ and conditioning lowers them by d_v and d_ℓ, the lesion becomes the maximum when $d_v - d_\ell > m(x_v) - m(x_\ell)$, regardless of whether the crop decision changes.

Robustness. The localization gain does not depend on the choice of readout or on refitting. In a post hoc analysis, fixed subtraction compared with its own separately tuned rotation gains $+9.66$ pp [+8.24,+11.38] in localized sensi-

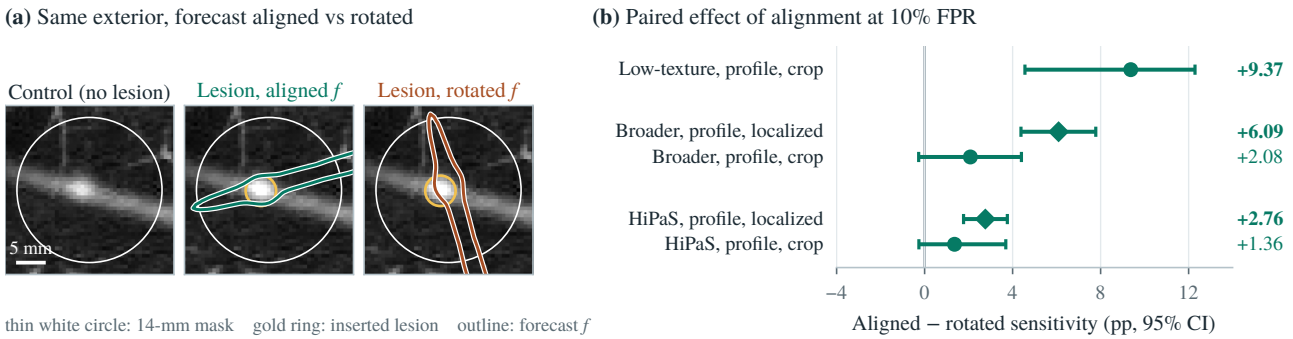


Figure 2. **The same forecast helps only when it is correctly placed.** (a) Development example showing the control (left) and its lesion-bearing counterpart with the aligned (middle) and rotated (right) forecast outline. The thin white circle is the 14-mm mask and the gold ring the inserted lesion; the control already exhibits a compact vessel response near the insertion site. All panels share one display window, and returned locations are not shown. (b) Aligned minus rotated sensitivity of the profile readout at the matched 10% FPR. Circles denote crop sensitivity and diamonds localized sensitivity; bold values indicate intervals that exclude zero. In all figures, intervals are 95% paired block-bootstrap CIs (3,000 draws of patients, or scans for HiPaS).

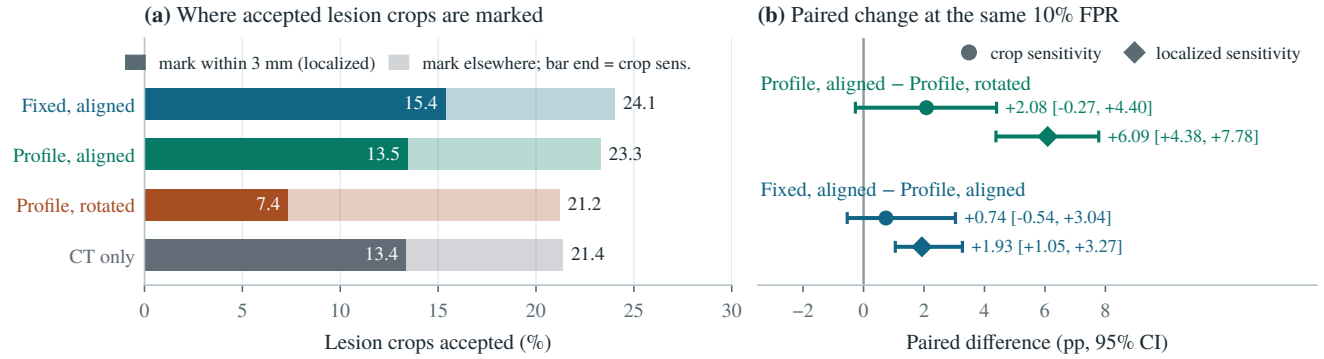


Figure 3. **Acceptance and localization are distinct decisions** (broader held-out set, matched 10% FPR). (a) Accepted lesion crops, divided into those whose returned location lies within 3 mm of the insertion (dark; localized sensitivity) and elsewhere (light). The bar end gives crop sensitivity. All scorers use the same search and the same lesion crops. (b) Paired differences in crop (circles) and localized (diamonds) sensitivity with 95% CIs.

tivity on the broader set and +2.39 pp [+1.22,+3.28] on HiPaS. Rotating only the field under the aligned classifier yields +4.26 pp [+2.39,+5.83] and +0.51 pp [-0.15,+1.09], respectively.

5.3. Readout Flexibility and Learned Baselines

Additional flexibility in the vessel amplitude did not improve performance. On the low-texture subset, the profile trailed the Hessian baseline by 0.84 pp [-6.15,+2.43], whereas fixed subtraction exceeded it by 3.41 pp [-0.23,+6.05]. On the broader set, the profile trailed fixed subtraction by 0.74 pp [-3.04,+0.54] in crop sensitivity, and fixed subtraction localized 1.93 pp [+1.05,+3.27] more insertions (15.38% of all insertions). On 18 development backgrounds with exact fields, where the forecast equals the added vessel, fixed subtraction and profiling both ordered 67.6% of insertion/control pairs correctly, compared with 62.0% for the plug-in readout. These results are consistent with the analysis in Sec. 3.3: a gain estimated inside the core can absorb part of the lesion.

The CT-only CNN reached 67.54% development AUROC and 36.28% sensitivity at the 10% cap, comparable to the ridge classifier on the CT-only readout (68.08% and 36.47%). On the low-texture subset its aligned-minus-rotated difference was −0.54 pp [-2.80,+0.62], and the residual input changed sensitivity by +2.16 pp [-1.19,+4.44] (Table 3). The network is not under-trained: it fits small training sets almost perfectly, and its development AUROC increases with the number of fitting patients (Fig. 4; Sec. S6). A classifier that pools learned features can therefore use the same field without exploiting its placement; the placement gain is specific to a readout in which the field can suppress a competing peak.

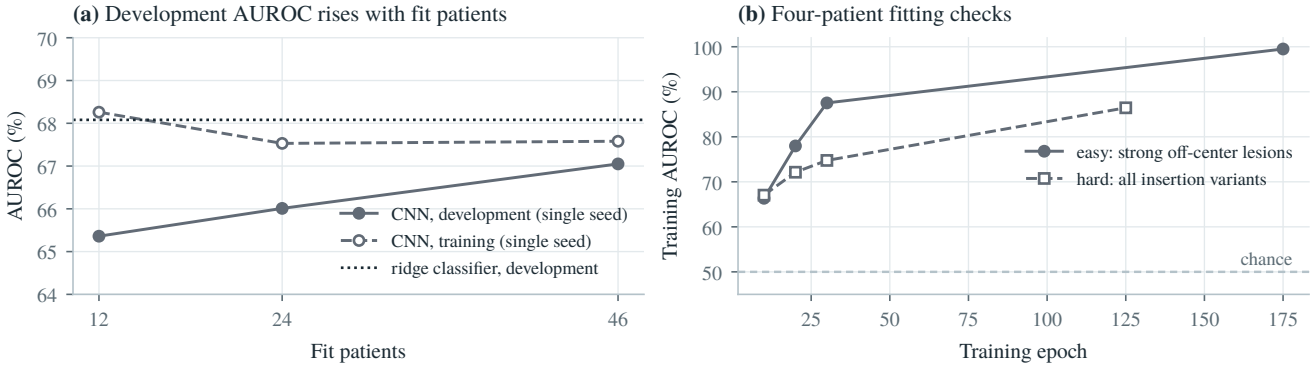


Figure 4. **Training behavior of the CT-only CNN.** (a) AUROC versus the number of fitting patients (single-seed snapshots; the reported 67.54% averages two seeds), with the ridge classifier on development for reference. (b) Training AUROC on four fitting patients for an easy setting (strong off-center lesions) and a hard setting (all insertion variants), the latter recorded up to 125 epochs.

Second input	Epoch	Dev. AUROC	AUROC	Sens.
CT only (smoothed CT)	30	67.54	68.78	34.28
Aligned forecast	20	67.54	68.81	35.14
Rotated forecast	30	67.81	69.13	35.68
Residual $y - f$	30	68.16	69.46	36.44

Table 3. **Hybrid CNN with four second inputs** (%). Architecture, data, seeds and budget are identical; the epoch is selected on development. AUROC and sensitivity (10% FPR cap) are reported on the low-texture subset, where the ridge classifier on the CT-only readout reaches 69.17% and 32.88%.

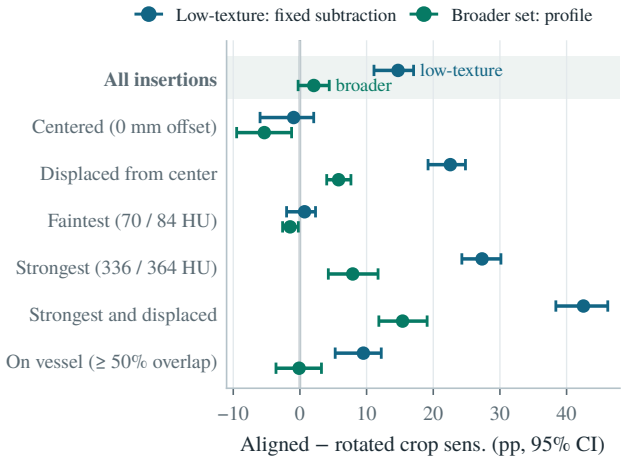


Figure 5. **Placement gains by insertion condition.** Aligned minus rotated crop sensitivity at each setting's global 10% FPR cap. Blue: fixed subtraction on the low-texture subset (development); green: the prespecified profile on the broader held-out set. Contrast values are given as low-texture / broader. Rows are not mutually exclusive.

5.4. Analysis by Insertion Condition

Fig. 5 decomposes the paired crop-level gain by insertion condition, evaluated at each setting's global cutoff. Gains concentrate where a vessel is most likely to produce the search maximum, namely for displaced, high-contrast insertions, which gained $+15.40$ pp [+11.84,+19.08] on the broader set. Centered insertions changed by -5.34 pp [-9.49,-1.25] and the faintest insertions by -1.47 pp [-2.60,-0.24], while insertions overlapping a vessel by at least 50% were unaffected (-0.12 pp [-3.59,+3.23]). A centered lesion is often already the strongest response, and a faint lesion leaves a weak residual even when the vessel is well explained. The low-texture subset exhibits the same pattern with larger effects.

We also evaluated a forecast-dominance score that is available before scoring, defined as the peak forecast relative to the texture of the shell (Sec. S5). Crops in the upper half of this score, 40.7% of the broader set, show a gain of $+3.29$ pp [+0.04,+6.03], but the high-minus-low interaction of $+2.03$ pp [-1.62,+5.11] is not resolved. Within the high-dominance band, alignment also raises the frozen FPR (11.19% *vs*. 9.39%; Sec. S5), so the score describes where alignment matters but does not by itself identify a better operating point.

5.5. Transfer to Native Vessels

On the 72 HiPaS scans, 7,713 of the 7,776 insertions lie within 2 mm of an annotated artery or vein. At matched FPR, alignment improved localized sensitivity by $+2.76$ pp [+1.77,+3.76] and crop sensitivity by $+1.36$ pp [-0.26,+3.69]. The localization gain therefore persists when real anatomy, including curved vessels that a straight chord approximates poorly, supplies the competing response. Both aligned readouts localize roughly twice as many insertions as the rotation (Fig. 6a), with 284 insertions newly localized and 69 lost.

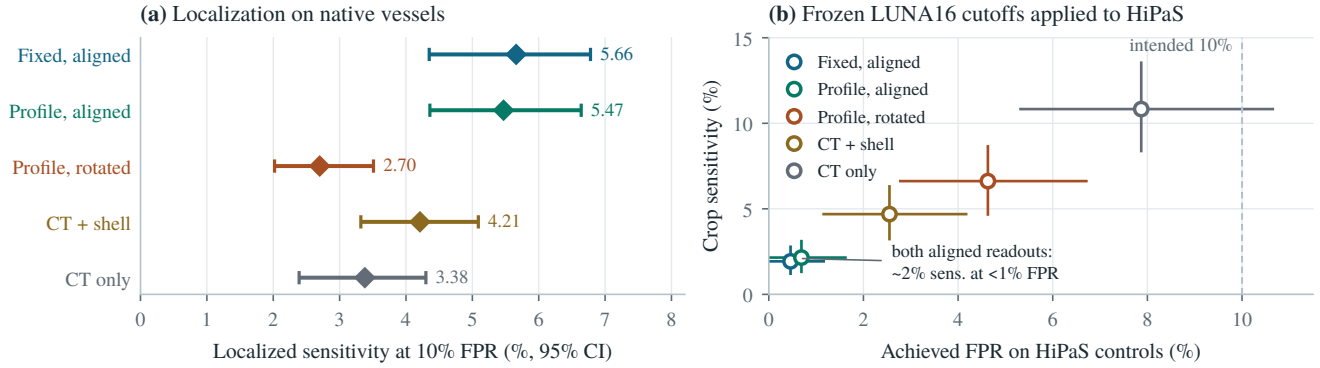


Figure 6. **Localization gains transfer to annotated native vessels; thresholds do not.** (a) Localized sensitivity on 7,776 HiPaS insertions at the within-HiPaS 10% FPR cap. (b) Crop sensitivity and achieved FPR when the frozen LUNA16 cutoffs are applied unchanged (intervals conditional on the cutoffs). Colors follow Fig. 3, with CT + shell in olive; diamonds denote localized sensitivity and open circles crop sensitivity at the frozen cutoffs, as in Fig. 7.

	Matched 10% FPR		LUNA16 cutoffs
Readout	sens.	loc. sens.	sens. / FPR
Fixed, aligned	16.26	5.66	1.93 / 0.46
Fixed, rotated	14.30	3.27	5.00 / 2.55
Plug-in, aligned	15.64	5.31	2.31 / 0.69
Plug-in, rotated	14.63	2.84	6.53 / 3.94
Profile, aligned	**15.52**	**5.47**	**2.15 / 0.69**
Profile, rotated	14.16	2.70	6.62 / 4.63
Profile, rot., aligned clf.	14.97	4.95	0.91 / 0.23
CT + shell	14.45	4.21	4.69 / 2.55
CT only	14.47	3.38	10.83 / 7.87
Hessian subtraction	11.33	–	12.04 / 10.88

Table 4. **HiPaS native vessels** (432 controls, 7,776 insertions; %). The matched columns use the within-HiPaS 10% FPR cap; the last column applies the independent LUNA16 cutoffs. Controls contain native anatomy without an inserted target.

Contrast	Intervals	Sensitivity Δ	FPR Δ
Broader LUNA16			
Lead−rot.	frozen	+4.50 [+2.62, +6.32]	+1.76 [−0.59, +4.12]
	recal.	+4.50 [−0.59, +8.31]	+1.76 [−2.35, +5.00]
Lead−fixed	frozen	−0.95 [−1.85, −0.10]	+0.00 [−1.32, +1.32]
	recal.	−0.95 [−3.80, +1.32]	+0.00 [−2.21, +1.76]
HiPaS			
Lead−rot.	frozen	−4.48 [−6.51, −2.58]	−3.94 [−6.02, −2.08]
	recal.	−4.48 [−7.09, −2.60]	−3.94 [−6.02, −1.62]
Lead−fixed	frozen	+0.22 [−0.05, +0.62]	+0.23 [+0.00, +0.69]
	recal.	+0.22 [−0.32, +0.73]	+0.23 [+0.00, +0.69]

Table 5. **Paired differences at the frozen LUNA16 cutoffs** (pp, 95% CI). Frozen intervals are conditional on the calibration set; recalibrated (recal.) intervals resample both calibration and evaluation blocks. The lead scorer is the aligned profile.

5.6. Ranking versus Calibration

When applied unchanged to HiPaS, the LUNA16 cutoffs yielded 2.15% sensitivity at 0.69% FPR for the aligned profile, compared with 6.62% at 4.63% for its rotation (Fig. 6b). Most scorers accepted far fewer HiPaS controls than the nominal 10%, so their realized operating points shifted. Fig. 7 shows the ROC curves underlying these operating points. On the broader set, both aligned readouts lie above the rotation and the CT-only scorer between approximately 3% and 25% FPR, so the matched result does not depend on the choice of a 10% cap. On HiPaS the crop-level curves nearly coincide, and the gain is confined to localization. The frozen LUNA16 cutoffs (open circles) fall close to the 10% cap on LUNA16 but far below it on HiPaS. Table 5 quantifies the contribution of calibration uncertainty: resampling the 175 calibration patients widens the broader lead-minus-rotation interval from $[+2.62, +6.32]$ to $[-0.59, +8.31]$ pp. On HiPaS the lead scorer accepts fewer insertions and fewer controls than its rotation, which reflects a shift of the operating point rather than a loss in ranking (Table 4).

The same distinction appears in two additional transfer conditions of the low-texture design, a LUNA16 morphology shift and an independent LNDb acquisition with shifted insertions [10]. Neither met its full transfer criteria at frozen cutoffs (Table 6); on the low-texture subset itself, the upper limit on excess FPR relative to the CT-only scorer was 4.92 pp against a prespecified 4 pp. Within each test set, however, the profile and Hessian sensitivities at 10% FPR were 37.93% and 37.44% under the morphology shift and 33.15% and 32.94% on LNDb, so ranking transferred better than the cutoffs. On LNDb, adding the spatial features to the scores of a published detector changed mean FROC by 0.00 pp [-0.21,+0.23] relative to the full CT-only classifier and by −0.13 pp [-0.33,+0.35] relative to the sham.

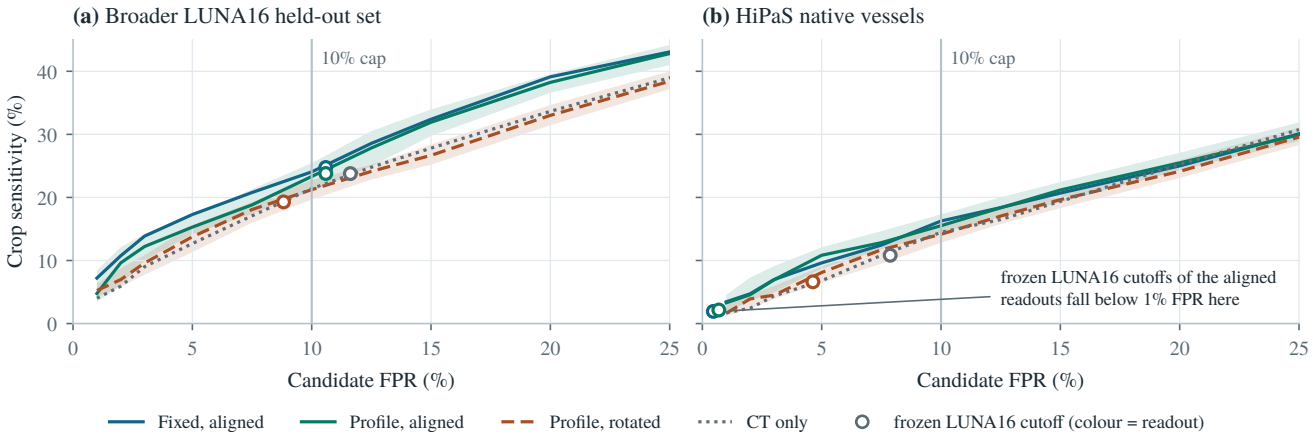


Figure 7. **ROC curves and transferred thresholds.** Curves use within-cohort negatives; shading gives pointwise 95% block intervals for the two profile placements. Open circles mark the frozen LUNA16 cutoffs, applied unchanged in both domains.

	Profile		Rotation		Hessian	
Condition	Sens.	FPR	Sens.	FPR	Sens.	FPR
Low-texture subset	46.85	10.38	37.19	10.02	38.39	5.46
Morphology shift	41.79	12.05	32.69	9.24	32.06	6.43
LNDb + morphology	28.04	6.49	29.27	9.14	36.09	12.09

Table 6. **Frozen operating points of the low-texture design** (%), using its own 276 LUNA16 calibration negatives, for the subset itself and two transfer conditions.

6. Discussion and Limitations

Implications for candidate scoring. A vessel forecast can appear useful for several reasons, since its shape and amplitude correlate with the type of background in a crop. The rotation sham, the exterior-summary control and the shared readouts together isolate the contribution of its *alignment*. Gains attributed to geometry should therefore be compared against a value-preserving placement sham, and fixed subtraction with the same search provides a competitive baseline for more flexible nuisance models. When a pipeline presents a returned location to a reader, moving that location onto the lesion is valuable even if the number of accepted crops barely changes; conversely, a gain in crop sensitivity can overstate the benefit when the returned location remains on the vessel.

Operating points. A matched-FPR effect establishes ranking separation at a given budget, whereas a shift in the negative score distribution changes the realized budget at an imported threshold. In our experiments, ranking gains from alignment transferred across acquisitions more reliably than calibrated cutoffs, and a deployment-oriented claim would require negatives from the target acquisition. A reliability-based rule for using the forecast selectively did not outperform always using it (Sec. S6).

Scope. All positive targets are inserted signals. The broader set varies texture and morphology under a known intervention, and HiPaS adds an independently annotated native background. Real nodules, scan-level proposal pipelines, partial-volume effects and per-scan false-positive rates are beyond the scope of this study, and an adjudicated scan-level evaluation would be required to quantify clinical detection performance.

7. Conclusion

We introduced a paired CT intervention with a shared exterior that reveals when vascular evidence from outside a candidate region changes the decision inside it. A correctly placed vessel forecast suppresses a competing vessel peak and returns the inserted target at the same false-alarm budget, and this localization effect persists on manually annotated native vessels. Fixed subtraction combined with a compact-template search exploits the same alignment as effectively as joint amplitude estimation.

LLM Usage Statement

OpenAI Codex assisted with study design, implementation, experiment execution, numerical analysis, literature lookup, figure preparation, and manuscript drafting and editing. Claude (Anthropic) assisted with revising the manuscript text and redrawing the figures from the recorded measurements. The quantitative figures were generated programmatically from recorded measurements. The qualitative panels use public CT crops or algorithmically generated phantoms. The authors are responsible for validating the work and its interpretation.

References

[1] Gady Agam, Samuel G. Armato III, and Changhua Wu. Vessel tree reconstruction in thoracic CT scans with application to nodule detection. *IEEE Transactions on Medical Imaging*, 24(4):486–499, 2005. 2

[2] Samuel G. Armato, III, Geoffrey McLennan, Luc Bidaut, et al. The lung image database consortium (LIDC) and image database resource initiative (IDRI): A completed reference database of lung nodules on CT scans. *Medical Physics*, 38 (2):915–931, 2011. 4

[3] Jean-Paul Charbonnier, Kaman Chung, Ernst T. Scholten, Eva M. van Rikxoort, Colin Jacobs, Nicola Sverzellati, Mario Silva, Ugo Pastorino, Bram van Ginneken, and Francesco Ciompi. Automatic segmentation of the solid core and enclosed vessels in subsolid pulmonary nodules. *Scientific Reports*, 8:646, 2018. 2

[4] Yuetan Chu, Gongning Luo, Longxi Zhou, et al. Deep learning-driven pulmonary artery and vein segmentation reveals demography-associated vasculature anatomical differences. *Nature Communications*, 16:2262, 2025. 4

[5] Alejandro F. Frangi, Wiro J. Niessen, Koen L. Vincken, and Max A. Viergever. Multiscale vessel enhancement filtering. In *Medical Image Computing and Computer-Assisted Intervention*, pages 130–137, 1998. 2

[6] Hao Gong, Jeffrey F. Marsh, Jamison E. Thorne, Shuai Leng, Cynthia H. McCollough, Joel G. Fletcher, and Lifeng Yu. Deep-learning lesion and noise insertion for virtual clinical trial in chest CT. In *Medical Imaging 2021: Physics of Medical Imaging, Proc. SPIE*, 2021. 2

[7] Xiaomeng Gu, Weiyang Xie, Qiming Fang, Jun Zhao, and Qiang Li. The effect of pulmonary vessel suppression on computerized detection of nodules in chest CT scans. *Medical Physics*, 47(10):4917–4927, 2020. 2

[8] Kaiyan Li, Weimin Zhou, Hua Li, and Mark A. Anastasio. Assessing the impact of deep neural network-based image denoising on binary signal detection tasks. *IEEE Transactions on Medical Imaging*, 40(9):2295–2305, 2021. 2

[9] Qiang Li, Shusuke Sone, and Kunio Doi. Selective enhancement filters for nodules, vessels, and airway walls in two- and three-dimensional CT scans. *Medical Physics*, 30(8): 2040–2051, 2003. 2

[10] João Pedrosa, Guilherme Aresta, Carlos Ferreira, et al. LNDb challenge on automatic lung cancer patient management. *Medical Image Analysis*, 70:102027, 2021. 8

[11] Marthony Robins, Justin Solomon, Pooyan Sahbaee, Martin Sedlmair, Kingshuk Roy Choudhury, Aria Pezeshk, Berkman Sahiner, and Ehsan Samei. Techniques for virtual lung nodule insertion: volumetric and morphometric comparison of projection-based and image-based methods for quantitative CT. *Physics in Medicine and Biology*, 62(18):7280–7299, 2017. 2

[12] Yoshinobu Sato, Shin Nakajima, Nobuyuki Shiraga, Hideki Atsumi, Shigeyuki Yoshida, Thomas Koller, Guido Gerig, and Ron Kikinis. Three-dimensional multi-scale line filter for segmentation and visualization of curvilinear structures in medical images. *Medical Image Analysis*, 2(2):143–168, 1998. 2

[13] Louis L. Scharf and Benjamin Friedlander. Matched subspace detectors. *IEEE Transactions on Signal Processing*, 42(8): 2146–2157, 1994. 2

[14] Arnaud Arindra Adiyoso Setio, Francesco Ciompi, Geert Litjens, Paul Gerke, Colin Jacobs, Sarah J. van Riel, Mathilde Marie Winkler Wille, Matiullah Naqibullah, Clara I. Sanchez, and Bram van Ginneken. Pulmonary nodule detection in CT images: False positive reduction using multi-view convolutional networks. *IEEE Transactions on Medical Imaging*, 35 (5):1160–1169, 2016. 2

[15] Arnaud A. A. Setio, Alberto Traverso, Thomas de Bel, et al. Validation, comparison, and combination of algorithms for automatic detection of pulmonary nodules in computed tomography images: The LUNA16 challenge. *Medical Image Analysis*, 42:1–13, 2017. 4

[16] Dijia Wu, Le Lu, Jinbo Bi, Yoshihisa Shinagawa, Kim L. Boyer, Arun Krishnan, and Marcos Salganicoff. Stratified learning of local anatomical context for lung nodules in CT images. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 2791–2798, 2010. 2

[17] Zitong Yu, Md Ashequr Rahman, Richard Laforest, Thomas H. Schindler, Robert J. Gropler, Richard L. Wahl, Barry A. Siegel, and Abhinav K. Jha. Need for objective task-based evaluation of deep learning-based denoising methods: A study in the context of myocardial perfusion SPECT. *Medical Physics*, 50(7):4122–4137, 2023. 2